

Day_3_Outlier & Distribution Analysis

September 9, 2026

0.1 Day 3 - Outlier & Distribution Analysis

0.2 Objective

The objective of this task is to identify outliers and understand the distribution of numerical variables in the Customer Churn dataset using IQR,Box plots, and skewness.

```
[2]: import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel(
    "Day_4_Ensemble_Learning_Customer_Churn.xlsx",
    sheet_name="Customer Churn Data"
)

df
```

```
[2]:
```

	CustomerID	Age	Tenure_Months	MonthlyCharges	Contract	\
0	1001	25	3	65	Month-to-month	
1	1002	34	18	72	One year	
2	1003	45	36	85	Two year	
3	1004	29	7	55	Month-to-month	
4	1005	52	48	95	Two year	
5	1006	41	24	78	One year	
6	1007	23	2	50	Month-to-month	
7	1008	38	30	88	One year	
8	1009	49	42	92	Two year	
9	1010	31	12	60	Month-to-month	
10	1011	27	8	68	Month-to-month	
11	1012	56	60	105	Two year	
12	1013	44	28	82	One year	
13	1014	36	20	75	One year	
14	1015	30	15	58	Month-to-month	
15	1016	47	40	90	Two year	
16	1017	22	4	45	Month-to-month	
17	1018	39	32	80	One year	
18	1019	53	54	100	Two year	
19	1020	33	16	70	Month-to-month	

	InternetService	TechSupport	PaymentMethod	Churn
0	Fiber optic	No	Electronic check	Yes
1	DSL	Yes	Bank transfer	No
2	Fiber optic	Yes	Credit card	No
3	DSL	No	Electronic check	Yes
4	Fiber optic	Yes	Credit card	No
5	DSL	Yes	Bank transfer	No
6	No	No	Mailed check	Yes
7	Fiber optic	Yes	Electronic check	No
8	Fiber optic	Yes	Credit card	No
9	DSL	No	Mailed check	Yes
10	Fiber optic	No	Electronic check	Yes
11	No	Yes	Bank transfer	No
12	DSL	Yes	Credit card	No
13	Fiber optic	No	Electronic check	Yes
14	DSL	No	Mailed check	Yes
15	Fiber optic	Yes	Credit card	No
16	No	No	Mailed check	Yes
17	DSL	Yes	Bank transfer	No
18	Fiber optic	Yes	Credit card	No
19	DSL	No	Electronic check	Yes

0.3 1. Interquartile Range (IQR)

The Interquartile Range(IQR) measures the spread of the middle 50% of the data.

$IQR = Q3 - Q1$

The lower and upper bounds are calculated using:

Lower Bound = $Q1 - 1.5 * IQR$ Upper Bound = $Q3 + 1.5 * IQR$

Values Outeside these bounds are considered potential outliers.

```
[5]: Q1 = df["MonthlyCharges"].quantile(0.25)
      Q3 = df["MonthlyCharges"].quantile(0.75)

      IQR = Q3 - Q1

      lower_bound = Q1 - 1.5 * IQR
      upper_bound = Q3 + 1.5 * IQR

      print("Q1:",Q1)
      print("Q3:",Q3)
      print("IQR:",IQR)
      print("Lower Bound:", lower_bound)
      print("Upper Bound:", upper_bound)
```

Q1: 63.75

Q3: 88.5

IQR: 24.75
Lower Bound: 26.625
Upper Bound: 125.625

0.4 2. Detecting Outliers

Values below the lower bound or above the upper bound are identified as potential outliers.

```
[7]: outliers = df[
      (df["MonthlyCharges"] < lower_bound) |
      (df["MonthlyCharges"] > upper_bound)
    ]

outliers
```

```
[7]: Empty DataFrame
Columns: [CustomerID, Age, Tenure_Months, MonthlyCharges, Contract,
InternetService, TechSupport, PaymentMethod, Churn]
Index: []
```

0.5 3. Box Plot

A box plot is used to visualize the distribution of numerical data and identify potential outliers.

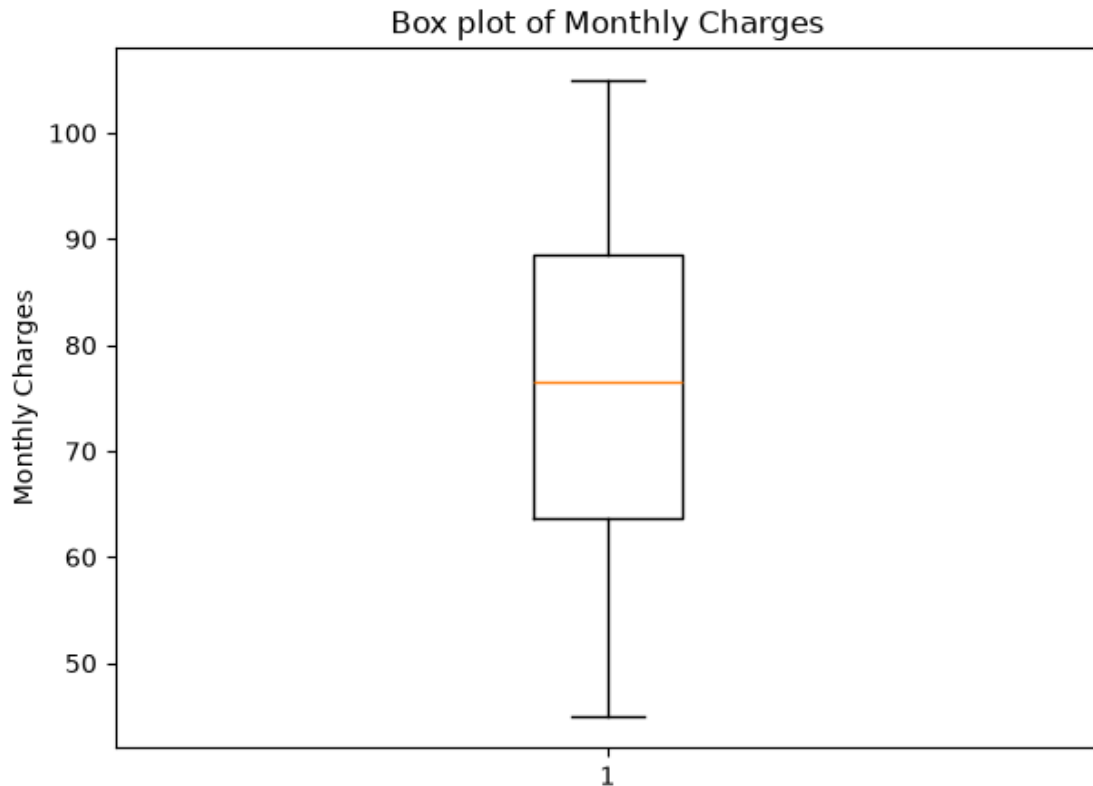
The box plot displays the median, quartiles, spread of the data, and potential outliers.

```
[9]: plt.figure(figsize=(7,5))

plt.boxplot(df["MonthlyCharges"])

plt.title("Box plot of Monthly Charges")
plt.ylabel("Monthly Charges")

plt.show()
```



0.6 4. Skewness

Skewness measures the asymmetry of the distribution of the numerical data.

- Skewness close to 0 indicates a nearly symmetric distribution.
- Positive skewness indicates that the distribution is skewed to the right.
- Negative skewness indicates that the distribution is skewed to the left.

```
[10]: df.skew(numeric_only=True)
```

```
[10]: CustomerID      0.000000  
Age                0.191145  
Tenure_Months      0.473062  
MonthlyCharges     -0.101752  
dtype: float64
```

0.7 5. Skewness of Monthly Charges

The skewness of MonthlyCharges is calculated to understand the shape of its distribution.

```
[13]: monthly_charges_skewness=df["MonthlyCharges"].skew()  
  
print("Skewness of MonthlyCharges:",monthly_charges_skewness)
```

Skewness of MonthlyCharges: -0.10175233554910668

0.8 Conclusion

In this task, Outlier and Distribution Analysis was performed on the Customer Churn dataset.

The IQR method was used to calculate Q1, Q3, IQR, and the lower and upper bounds for identifying potential outliers. A box plot was created to visualize the distribution and potential outliers of MonthlyCharges. Skewness was also calculated to understand the shape and asymmetry of the numerical data.

These techniques help in understanding the quality, spread, and distribution of the dataset before applying further data analysis or machine learning techniques.